

A Primer on Study Design, Analysis, and Interpretation

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Disclosures

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~~An In~~convenient Truth

Everything Now Gets Published

- Noise is/has quickly overtaken signal
- Difficult to discern robust vs. spurious findings
- Impactful research is nearly always high quality
- We should chase *impact* not mere publication

Most important elements of quality are determined *before* manuscript preparation

- Study Design
- Statistical Analysis Plan

- Manuscript Preparation

Ethical issues

- Obtain and remain current on local ethical guidance
- Study design considerations explicitly in the context of ethical implications
- Seek out the expertise of an ethicist as part of team

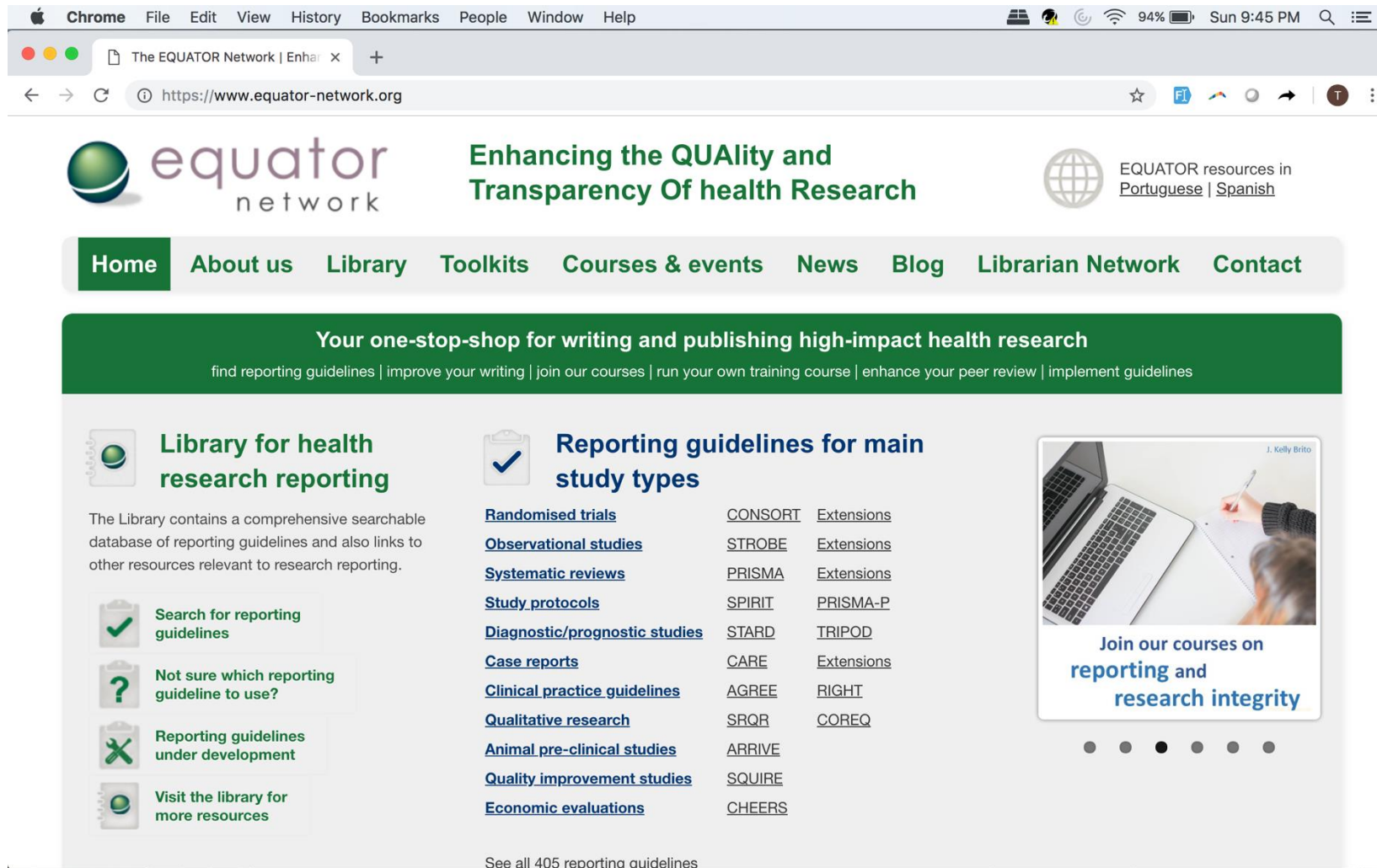
- Suspected ethical issues are cause for immediate rejection
 - Clear statements about oversight and approval
 - International standards vary; be transparent about local law
- Appropriate consent/assent is paramount
 - Often poorly reported

Study design

- From clear hypotheses or objectives, identify an appropriate study design
 - Not the other way around
- Each study design represents a world of considerations
 - Case-control vs cohort
- Consider pre-registration
 - PROSPERO, Clinicaltrials.gov
- Seek out an expert of the design of the study, not just the content

- Name the design
- Craft the methods from the point-of-view of the study *design*
 - Unique methods for each
- Avoid accidental blending of design features
 - Cohort study with prediction model

Report with enough detail to allow replication



The screenshot shows the EQUATOR Network website in a Chrome browser. The browser's address bar displays the URL <https://www.equator-network.org>. The website's header features the EQUATOR Network logo, the tagline "Enhancing the QUALity and Transparency Of health Research", and links to resources in Portuguese and Spanish. A navigation menu includes links to Home, About us, Library, Toolkits, Courses & events, News, Blog, Librarian Network, and Contact. A green banner below the navigation menu states: "Your one-stop-shop for writing and publishing high-impact health research" and lists services: "find reporting guidelines | improve your writing | join our courses | run your own training course | enhance your peer review | implement guidelines". The main content area is divided into three sections. The left section, "Library for health research reporting", describes a searchable database of reporting guidelines and includes four links: "Search for reporting guidelines", "Not sure which reporting guideline to use?", "Reporting guidelines under development", and "Visit the library for more resources". The middle section, "Reporting guidelines for main study types", lists various guidelines in two columns: Randomised trials, Observational studies, Systematic reviews, Study protocols, Diagnostic/prognostic studies, Case reports, Clinical practice guidelines, Qualitative research, Animal pre-clinical studies, Quality improvement studies, Economic evaluations, CONSORT, STROBE, PRISMA, SPIRIT, STARD, CARE, AGREE, SRQR, ARRIVE, SQUIRE, and CHEERS. Each guideline name is followed by a link to "Extensions". The right section features a photo of a person writing on a notepad next to a laptop, with the text "Join our courses on reporting and research integrity". At the bottom of the page, a link says "See all 405 reporting guidelines".

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The EQUATOR Network | Enhance x

<https://www.equator-network.org>

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**Enhancing the QUALity and
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Your one-stop-shop for writing and publishing high-impact health research

find reporting guidelines | improve your writing | join our courses | run your own training course | enhance your peer review | implement guidelines

 **Library for health
research reporting**

The Library contains a comprehensive searchable database of reporting guidelines and also links to other resources relevant to research reporting.

-  **Search for reporting guidelines**
-  **Not sure which reporting guideline to use?**
-  **Reporting guidelines under development**
-  **Visit the library for more resources**

 **Reporting guidelines for main
study types**

Randomised trials	CONSORT	Extensions
Observational studies	STROBE	Extensions
Systematic reviews	PRISMA	Extensions
Study protocols	SPIRIT	PRISMA-P
Diagnostic/prognostic studies	STARD	TRIPOD
Case reports	CARE	Extensions
Clinical practice guidelines	AGREE	RIGHT
Qualitative research	SRQR	COREQ
Animal pre-clinical studies	ARRIVE	
Quality improvement studies	SQUIRE	
Economic evaluations	CHEERS	

 **Join our courses on
reporting and
research integrity**

See all 405 reporting guidelines

Statistical analysis plan (SAP)

- Have an SAP!
- Develop SAP in conjunction with the design and before looking at the data
 - 1 paragraph to 30 pages
- Develop sample size considerations
- Seek out statistical support with expertise with the study design
 - Clinical trialist vs. Meta-analyst

- Report the SAP in sufficient detail to allow replication
 - Distinguish between planned and post hoc analyses
- Be explicitly conscious of the objectives of the analysis
 - Description, inference (explanation), prediction
- Strongly consider providing data and code underlying the analysis

Avoid *THE* stats section

Statistical analysis

Categorical variables are reported as count (%) and continuous variables as mean (SD) or median (IQR). The presence of normal distribution was verified by Kolmogorov-Smirnoff test. We used the *t* test to assess differences between parametric continuous variables, the Mann-Whitney *U* test for non-parametric variables, the χ^2 test for categorical variables, and the Fisher exact test for 2×2 tables. No correction for multiple testing was done. A two-sided $p < 0.05$ was considered statistically significant. All analyses were done with SPSS version 24.0.

Results reporting

- Plan the results reporting framework *before* conducting the analysis
 - Table shells help prevent unconscious bias in reporting
- SAP should determine the estimand (the thing being estimated)
 - Which sample?
 - Which covariates
 - Missing data?
 - Outliers?

- Craft the results from the point-of-view of the study *design*
 - Table 1 stratification
 - Sources of confounding or bias
- No matter the design, extensive use of descriptive statistics
- Move beyond null hypothesis testing to communication of uncertainty
 - Measurement error
 - Confidence/Plausibility/Credibility interval

Interpretation

- Most elements of interpretation pre-specified in SAP
 - NOT based on sifting through results
 - Surprising/exploratory findings require appropriate considerations
- Minimal clinically important difference (MCID)
- Design a study that would change belief or practice
- Based on proper use of inferential methods
 - Null hypothesis testing

- Interpretations based on predefined considerations
 - Avoid promoting secondary outcomes to primary importance
 - Avoid post-hoc subgroups
- Interpretations based on updating beliefs in the context of previous information
 - Surprising findings viewed with caution
 - Overstating findings should be anathema
- Avoid 'spin', aggrandizement, or inappropriate bromides
 - "This first of its kind..."
 - "... Safe and effective ..."

Welcome Statistical Editors!



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Thank You!

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